

DOWNSTREAM PORT

J1
USB3_A

VBUS 1 5V

D- 2 USB_P1_D-

D+ 3 USB_P1_D+

SSRX- 5 USB_P1_RX1_N

SSRX+ 6 USB_P1_RX1_P

8 USB_P1_TX1_N_1

9 USB_P1_TX1_P_1

10 SHIELD

7 DRAIN

4 GND

GND

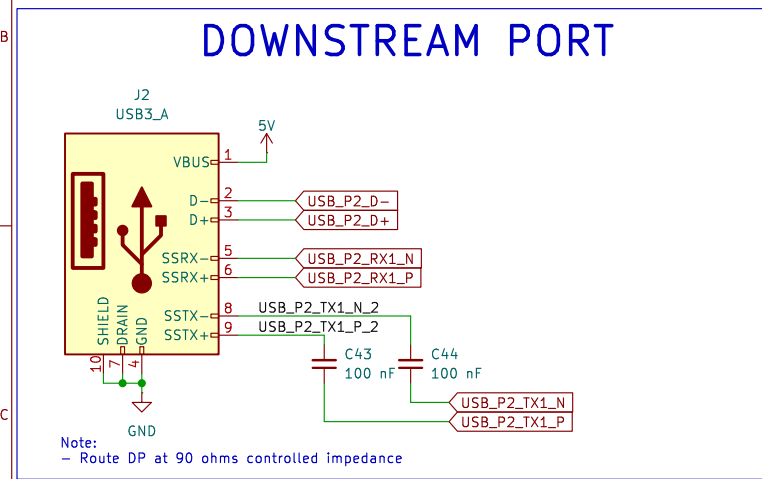
C52 100 nF

C53 100 nF

USB_P1_TX1_N

USB_P1_TX1_P

Note:
- Route DP at 90 ohms controlled impedance



DOWNSTREAM PORT

J3
USB3_A

5V

VBUS 1

D- 2

D+ 3

SSRX- 5

SSRX+ 6

SSTX- 8

SSTX+ 9

SHIELD 7

DRAIN 4

GND

USB_P3_D-

USB_P3_D+

USB_P3_RX1_N

USB_P3_RX1_P

USB_P3_TX1_N

USB_P3_TX1_P

C45 100 nF

C46 100 nF

Note:

- Route DP at 90 ohms controlled impedance

DOWNSTREAM PORT

The diagram illustrates the pin connections for a USB downstream port connector J4 (USB3_A). The connector has 9 pins, numbered 1 through 9. The connections are as follows:

- Pin 1: VBUS, connected to 5V.
- Pin 2: D-, connected to USB_P4_D-.
- Pin 3: D+, connected to USB_P4_D+.
- Pin 5: SSRX-, connected to USB_P4_RX1_N.
- Pin 6: SSRX+, connected to USB_P4_RX1_P.
- Pin 8: SSTX-, connected to USB_P4_TX1_N_4.
- Pin 9: SSTX+, connected to USB_P4_TX1_P_4.

Additionally, there are two capacitors, C47 and C48, connected in parallel between the TX1_P and TX1_N lines (pins 8 and 9). Both capacitors have a value of 100 nF.

Note:
- Route DP at 90 ohms controlled impedance

UPSTREAM PORT

The schematic diagram illustrates the USB Upstream Port circuit. It features a USB Type-C connector (J5, TYPE-C24P_QT_143) connected to a USB controller (U3, HD3SS32201RNHT). The USB controller is connected to a USB hub (U3, HD3SS32201RNHT). The schematic shows various signal lines, power planes (VBUS, GND), and components like capacitors (C37, C38, C49, C50, C54, C55) and resistors (R17, R18, R19, R9). It also shows the connection to a USB Type-C connector (J5) and a USB Type-A connector (J6).

Components and Connections:

- USB Type-C Connector (J5):** TYPE-C24P_QT_143. Pins include VBUS_A, S5TXP1, S5TXN1, CC1, DP1, DN1, S5BU1, S5SRXN2, S5SRXP2, GND_A, SHIELD, VBUS_B, S5SRXP1, S5SRXN1, S5BU2, DN2, DP2, CC2, S5TXN2, S5TXP2, GND_B, B4, B11, B10, B8, B7, B6, B5, B3, B2, B1.
- USB Controller (U3):** HD3SS32201RNHT. Pins include ADDR, CURRENT_MODE, ENN_LCC, ENN_MUX, PORT, VBUS_DET, CC1, CC2, RX1N, RX1P, RX2N, RX2P, RXN, RXP, SCL/OUT2, SDA/OUT1, TX1N, TX1P, TX2N, TX2P, TXN, TXP, EXP_PAD, GND.
- Capacitors:** C37 (100 nF), C38 (100 nF), C49 (100 nF), C50 (100 nF), C54 (100 nF), C55 (100 nF).
- Resistors:** R17 (4.7 KΩ), R18 (1 MΩ), R19 (4.7 KΩ), R9 (200 KΩ).
- Connectors:** USB_P0_TX1_P, USB_P0_TX1_N, USB_P0_RX1_P, USB_P0_RX1_N, USB_P0_D+, USB_P0_D-, USB_P0_RX2_P, USB_P0_RX2_N, USB_P0_TX2_P, USB_P0_TX2_N, USB_UP_P0_RX1_N, USB_UP_P0_RX1_P.
- Power and Ground:** VBUS, GND, 3V3.

Note:
 - Route DP at 90 ohms controlled impedance

The diagram illustrates a 12V to 5V step-down voltage regulator circuit. The main component is the TPS54331DDAR IC (U2), which is configured as a buck converter. The input voltage (VIN_12V) is connected to the VIN pin (pin 2) through a 22 µF capacitor (C8). The EN pin (pin 3) is connected to GND through a 10 µF capacitor (C9). The SS pin (pin 4) is connected to GND through a 10 nF capacitor (C6). The VSENSE pin (pin 5) is connected to the PH pin (pin 8) through a 100 nF capacitor (C7). The PH pin (pin 8) is connected to the SW pin (pin 9) through a 49.9 kΩ resistor (R3). The SW pin (pin 9) is connected to the GND pin (pin 7) through a 4.7 nF capacitor (C4). The GND pin (pin 7) is connected to GND through a 47 pF capacitor (C3). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 100 nF capacitor (C5). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 6.8 µH inductor (L1). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 22 µF capacitor (C1). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 10 µF capacitor (C10). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 22 µF capacitor (C2). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 10 µF capacitor (C11). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 100 nF capacitor (C12). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 10 kΩ resistor (R1). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 1.8 kΩ resistor (R2). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 100 Ω resistor (R4). The SW pin (pin 9) is connected to the SW pin (pin 9) through a 5V output. The SW pin (pin 9) is connected to the SW pin (pin 9) through a VSENSE output. The SW pin (pin 9) is connected to the SW pin (pin 9) through a GND output.

Components:

- U2: TPS54331DDAR
- C8: 22 µF
- C9: 10 µF
- C7: 100 nF
- C6: 10 nF
- C5: 100 nF
- C4: 4.7 nF
- C3: 47 pF
- C1: 22 µF
- C10: 10 µF
- C2: 22 µF
- C11: 10 µF
- C12: 100 nF
- R3: 49.9 kΩ
- R1: 10 kΩ
- R2: 1.8 kΩ
- R4: 100 Ω
- L1: 6.8 µH
- D1: SS54
- D3: D_Zener
- Q1: AO3401A
- F1: Polyfuse_Small
- J6: DC-005

Notes:

- Barrel Jack and Reverse Voltage & Current Protection
- 12V Input Voltage

Figure 1: Recommended capacitor placement for the GL3510

5V

C16 22 uF, C14 100 nF

3V3

C17 100 nF, C18 100 nF, C19 100 nF, C20 100 nF, C21 100 nF, C22 100 nF, C23 100 nF

VBUS

C36 10 uF

5V

C39 10 uF, C40 10 uF, C41 10 uF, C42 10 uF

DVDD_1V2

C24 100 nF, C25 100 nF, C26 100 nF, C51 100 nF

AVDD_1V2

C27 100 nF, C28 100 nF, C29 100 nF, C30 100 nF, C31 100 nF, C32 100 nF

3V3

C59 100 nF

VBUS

C60 100 nF

Note:

- Place caps closer to GL3510 chip pin
- Caps placement should be largest to smallest value

Note:

- Place C36 capacitor closer to Upstream Port VBUS pin
- Place C39, C40, C41, C42 caps closer to Downstream Port VBUS Pin

Note:

- Place caps closer to GL3510 chip pin
- Caps placement should be largest to smallest value

Note:

- Place C59 capacitor closer to MUX VCC33 pin
- Place C60 capacitor closer to MUX VDD5 pin

Sheet: /			
File: USB 3.1 Hub.kicad_sch			
Title: SUPERSPEED USB HUB			
Size: A3	Date: 2026-03-05	Rev: V1	
KiCad 5.0.0		Id: 1/1	

Size: A3	Date: 2026-03-05	Rev: V1
KiCad EDA 9.0.0		Id: 1/1